

Subcontracting Desk

From informal chat decisions
to verified, accountable outsourcing dispatch

Group NN

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THE CHAT MESSAGE

"Send it to QuickStitch, need it back by the 19th."



THE DESK'S ANSWER

GiantWeave

back Fri 10 Apr, on time · Sat 11 Apr if reworked

QuickStitch not considered: excluded by Boss
Every number traced to a tool output

A batch goes outside on one chat message, unchecked and unrecorded

TODAY

"@Chen — order 042, 1,500 crewnecks. Send it to QuickStitch, need it back by the 19th."

1

Can they make it? Not every workshop makes every product.

2

Can they take it now? A fast shop with a 3-day queue is not fast today.

3

Are they allowed to? One shop is suspended after an audit; one is on trial, small batches only.

Get one wrong and the batch bounces back a day later. In a ten-day cycle that is the margin gone.

The decision: which eligible workshop, with capacity, returns the batch by its due date — and why.

WHAT THE DESK MUST DO

Read the chat. Extract the request; ask when it is ambiguous or incomplete; refuse, with a reason and an alternative, when the named workshop cannot take it; honour constraints and record who set them.

Decide on numbers. Keep the workshop register; estimate a delivery date per candidate before committing; allocate with the reasoning attached; when nobody can make the date, say so and lay out the options.

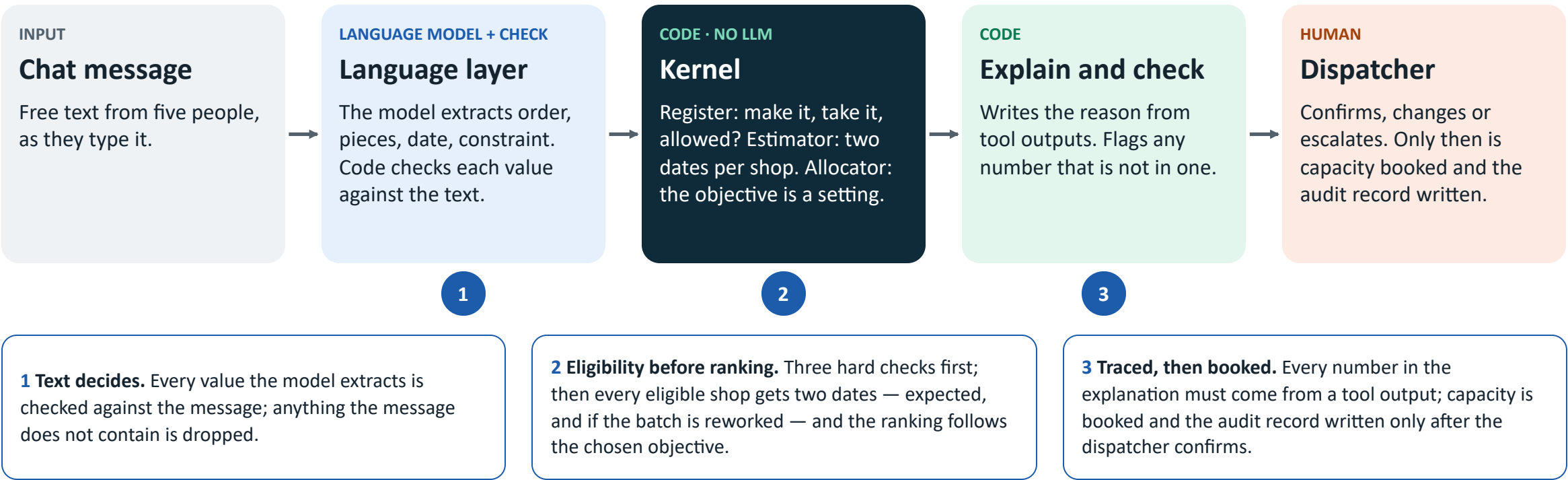
Leave a record. An auditable record of every decision — what the group chat never produced.

Optional, assessed after Sprint 2: the stage extension (--stages). Not pursued: multi-agent negotiation — the mechanism is not graded.

BUSINESS VALUE

- **Dispatcher:** faster decisions with verified recommendations
- **Factory:** fewer late deliveries, shown in simulation
- **Customer:** more reliable delivery-date commitments

The model reads, code decides, the dispatcher confirms



ONE REAL MESSAGE, R12, THROUGH THE FIVE STAGES

“ORD-053 — 800 polo shirts for UrbanThread, due Apr 10. Boss says nothing new to QuickStitch this week.”

ORD-053 · 800 pcs · due 10 Apr · constraint: QuickStitch excluded — standing, set by Boss

4 shops eligible. GiantWeave back Fri 10 Apr; Sat 11 Apr if reworked, still on time

“GiantWeave: on time; it only makes tops, so the two shops that also make accessories stay free for tighter batches.”

Confirmed → ledger updated, audit record A0004

Primary objective: minimise late deliveries

Late-delivery rate = batches returned after their due date ÷ all batches. Also reported: mean days late (0 when on time). Both are the simulator's own metrics.

WHY THIS ONE

- 1 Relevant.** The due date is the one constraint every order carries; quick response exists to protect it — “the factory that farms out a batch today beats the one that decides on Thursday.”
- 2 Measurable.** The simulator records when every batch returns, so lateness is computed, not estimated.
- 3 Controllable.** Lateness depends on queue, capacity and transport — exactly what the choice of workshop changes.

The three baselines are 34–92% late in simulation: the largest gap, but supporting evidence, not the reason.

THE OTHER THREE

Defects, fairness. Switchable settings of the same allocator; results reported on the same 120 orders.

Hybrid. One weighted setting (late 1.0, finish 0.3, defect 0.25, load 0.5). Metrics scaled 0–1 first; weights at the knee of the lateness–defect curve, checked on held-out seeds.

Robust to failure. Tested with the shock scenario; guardrail: lateness no worse than random.

Declared tie-break: when two settings are equally late, we take the shorter turnaround — the track names turnaround days as the business impact.

The loop runs end to end, and the first results are in

To confirm (13)Needs a decision (7)Ask requester (7)Can't answer (3)Done (0)

Confirm 7 safe recommendations in one click

Held back for you: 3 tight (a rework would make them late), listed first, and 3 that would take a workshop a tight one needs.

	ID	Order	Decision	Back	Spare
☐	R12	ORD-053	★ GiantWeave · tight	Fri 10 Apr	0 d
☐	R17	ORD-017	★ GiantWeave · tight	Fri 10 Apr	0 d
☐	R21	ORD-081	★ GiantWeave · tight	Sun 12 Apr	1 d
☐	R25	ORD-109	★ GiantWeave	Thu 09 Apr	4 d
☐	R09	ORD-045	★ Nimble Needle	Sat 04 Apr	5 d
☐	R06	ORD-096	★ Nimble Needle	Sat 11 Apr	6 d
☐	R16	ORD-024	★ FreshStart	Mon 06 Apr	9 d
☐	R04	ORD-041	★ Little Loom	Sat 11 Apr	10 d
☐	R22	ORD-040	★ GiantWeave	Sat 11 Apr	10 d
☐	R23	ORD-005	★ Little Loom	Fri 10 Apr	10 d
☐	R26	ORD-010	★ GiantWeave	Sat 11 Apr	14 d
☐	R27	ORD-095	★ SteadyHands	Fri 10 Apr	16 d
☐	R07	ORD-063	★ FreshStart	Sat 04 Apr	25 d

New message

Boss

e.g. ORD-066 — 1000 polo shirts, due Apr 20

Send to desk

R12 · 08:48 · Ravi

Confirm

ORD-053 — 800 polo shirts for UrbanThread, due Apr 10. Boss says nothing new to QuickStitch this week, they're swamped.

ORD-053 · 800 pcs TOPS · due Fri 10 Apr (9 days) · avoid QuickStitch · parsed by rules

★ GiantWeave · back Fri 10 Apr · on time, 0 days to spare · if reworked: Sat 11 Apr (late) · cost 880

Why: on time, though a rework would make it late (no eligible shop is safe against that); it only makes TOPS, so the shops that can take other work stay free for tighter batches.

☒ every number comes from a tool output · 4 tool calls

• session constraint recorded: exclude QuickStitch (by Boss)

Eligible workshops · ranked by lateness

	Workshop	Back	Days to spare	If reworked	Days	Queue	Wo
★	GiantWeave	Fri 10 Apr	0	Sat 11 Apr (late)	9.1	3	
	Nimble Needle	Thu 09 Apr	1	Sun 12 Apr (late)	8.3	1	€
	SteadyHands	Fri 10 Apr	0	Mon 13 Apr (late)	9.3	2.5	4
	BudgetWorks	Mon 13 Apr	-3	Tue 14 Apr (late)	11.6	5	3

Not eligible: QuickStitch (excluded by request) · Little Loom (cannot make TOPS (equipped for ACCESSORIES)) · OldMill (status=SUSPENDED) · FreshStart (batch of 800 exceeds its 300-piece limit)

Confirm GiantWeave

Nimble Needle · back Thu 09 Apr

Send to Nimble Needle

One message in, one recommendation with two dates and a reason, one click to confirm, one audit record.

IN SIMULATION · OFFICIAL 120 ORDERS · MEAN OF 10 SEEDS

Policy	Late batches	Mean days late
random (baseline)	34.2%	6.13
cheapest (baseline)	91.8%	81.9
earliest-finish (simple rule, no LLM)	1.7%	0.18
ours (lateness setting)	1.7%	0.09

On 200 fresh order books the simple rule never saw, ours is ahead: 0.9% vs 1.7% late on 100 jittered sets (better on 72, worse on 3). Under shock the simple rule still edges us, 1.9% vs 2.0%: one batch in ten runs. Source: simulate.py v3, commit 1c6cb61.

- **56 automated tests pass**; every number in every reply on the 30-message inbox traces to a tool output.
- **All six of us labelled the 30 messages independently** (κ 0.70); labels frozen on 7 Oct; counts per kind match the instructor's mix.
- **The language model was measured, then corrected**: GPT-5 mini now reads all 30 official and 38 new messages right; the cheapest model started at 86%.

What could go wrong, and what we are assuming

RISK	IMPACT	MITIGATION
No clean language test yet. The 30 official messages were used in development; the 38 new ones are AI-written.	We may overstate how well the model reads.	Held-out inbox: 44 messages written by four teammates who do not touch the prompt; answers frozen before the first run (frozen 14 Oct, first run 15 Oct).
The simulator is kinder than reality. Seven-day week; the estimator does not know when a workshop closes.	Lateness looks better than it would be.	Sunday variant reported separately (simple rule 1.7 → 3.7%, ours → 2.1%); closed-workshop detection in Sprint 2.
Our online policy is not globally optimal. A hindsight search that knows every order in advance reaches 0% late in the tested setting.	Room we have not captured.	A dispatcher cannot see future orders; we report the gap and keep the rule simple enough to explain.

Assumptions: requests arrive in time order and a confirmed dispatch occupies capacity · a standing constraint (“nothing new to QuickStitch this week”) applies to every later request · a reworked batch takes half as long again (simulator rule) · the data holds no customer prices and no history, so those questions are declined.

Sprint 2 (30 Oct) and Sprint 3 (13 Nov): from a working loop to evidence we can defend

WORK	ACCEPTANCE	OWNER	WHEN
Held-out inbox	44 hand-written messages, answers frozen first, ≥ 90% right behaviour	张锦若 freezes; 李岩, 吴若晗 run	written 9–11 Oct; frozen 14 Oct; first run 15 Oct
Closed-workshop detection	In the shock run the closed shop is detected and gets no new batches; shock lateness no worse than normal	吴若晗	design 14 Oct; done 27 Oct
User test	2–3 students from other groups complete the 9 tasks	李岩, 屈妍玥	by 28 Oct
Public demo	Reachable from an outside browser; one dispatch completed	李岩 (吴杰 logs in)	by 28 Oct
Evaluation draft	10 failure cases with root causes	李岩, 吴若晗 (causes); 吴杰, 许佩瑶 (text)	24–29 Oct
Optional: stage extension (--stages)	Assessed after Sprint 2, if time allows	—	—

SUCCESS CRITERIA · MEASURED ON v3 DATA

≤ 2%

late on the official orders, normal and shock
(now 1.7% / 2.0%)

Beat it

fewer late batches than the simple rule on
fresh order books (now 0.9% vs 1.7%)

≥ 90%

right behaviour on the held-out inbox, labels
frozen first

100%

numbers traced; every “if reworked” date kept
(now 100%; 99% in shock)

Appendix A - Results on v3 data, mean of 10 seeds

Policy	Late %	Late days	Mean days	Defective pcs %	Max share %	Cost	Late % in shock
random (official)	34.2	6.13	21.9	5.6	23	111,780	38.2
greedy_biggest (official)	84.2	26.53	50.3	5.3	70	108,430	84.9
cheapest (official)	91.8	81.92	106.2	8.2	70	85,960	91.8
earliest_finish (simple rule)	1.7	0.18	8.3	7.2	26	115,566	1.9
lateness: specialist-first rule (ours)	1.7	0.09	9.3	7.7	28	115,438	2.0
use_slack: latest safe finish (ours)	1.0	0.07	14.0	6.8	32	105,502	1.7
lateness, round 1 (ours)	1.7	0.18	8.2	7.5	26	115,592	1.9
hybrid (ours)	4.3	0.29	10.1	6.1	29	111,238	5.5
defects (ours)	28.7	2.23	21.1	5.1	43	110,396	31.0
fairness (ours)	6.8	0.64	12.6	7.1	20	113,088	10.1
cost (ours)	4.4	0.24	17.2	5.7	38	100,931	5.2

simulate.py v3, commit 1c6cb61. use_slack is late less often and costs less, but takes 14 days on average. Single-seed defect draws move defective pcs by about ± 2 points.

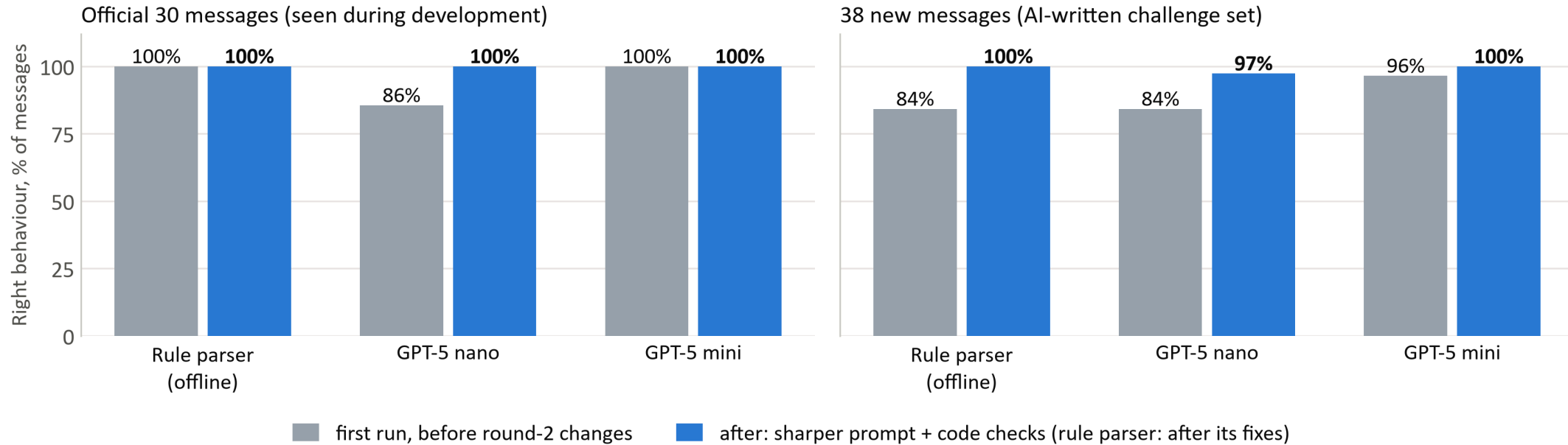
Appendix B · How the desk sorted the 30 messages

R01 ask: missing info	R02 can't answer	R03 ask: which order	R04 ★ allocate	R05 ask: missing info	R06 ★ allocate
R07 ★ allocate	R08 refuse	R09 allocate	R10 ask: missing info	R11 refuse	R12 allocate
R13 can't answer	R14 refuse	R15 ask: which order	R16 ★ allocate	R17 ★ allocate	R18 can't answer
R19 ask: which order	R20 no on-time option	R21 allocate	R22 ★ allocate	R23 ★ allocate	R24 refuse
R25 allocate	R26 ★ allocate	R27 ★ allocate	R28 no on-time option	R29 ask: which order	R30 no on-time option

■ allocate
■ no on-time option
■ ask: which order
■ ask: missing info
■ refuse
■ can't answer

★ changed by the instructor in v3. Shown with the morning queues; the counts by kind match the instructor's published mix. Processed in order, 11 of the 13 on-time requests stay on time (R21 is 1 day late; R25 is 1 day late); at most 12 can.

Appendix C · The language layer, measured on two inboxes



- **The rule parser was built around the 30:** 100% there, 84% on new wording at first try.
- **Model choice matters:** with the round-1 prompt GPT-5 nano gets 86% of the 30 right, GPT-5 mini 100%. After our checks nano still misreads one new phrasing (a date “sooner than planned, not said when”) in 3 runs of 3.
- **Not a clean test yet:** the 38 new messages are AI-written and two of our fixes came from errors seen on them. The held-out inbox will be written by teammates.

Appendix D · Promised dates, and a six-day week

QUOTING TWO DATES · 1,200 BATCHES, LATENESS RULE

94%

of expected dates hold. Every miss in normal runs is a defect rework.

100%

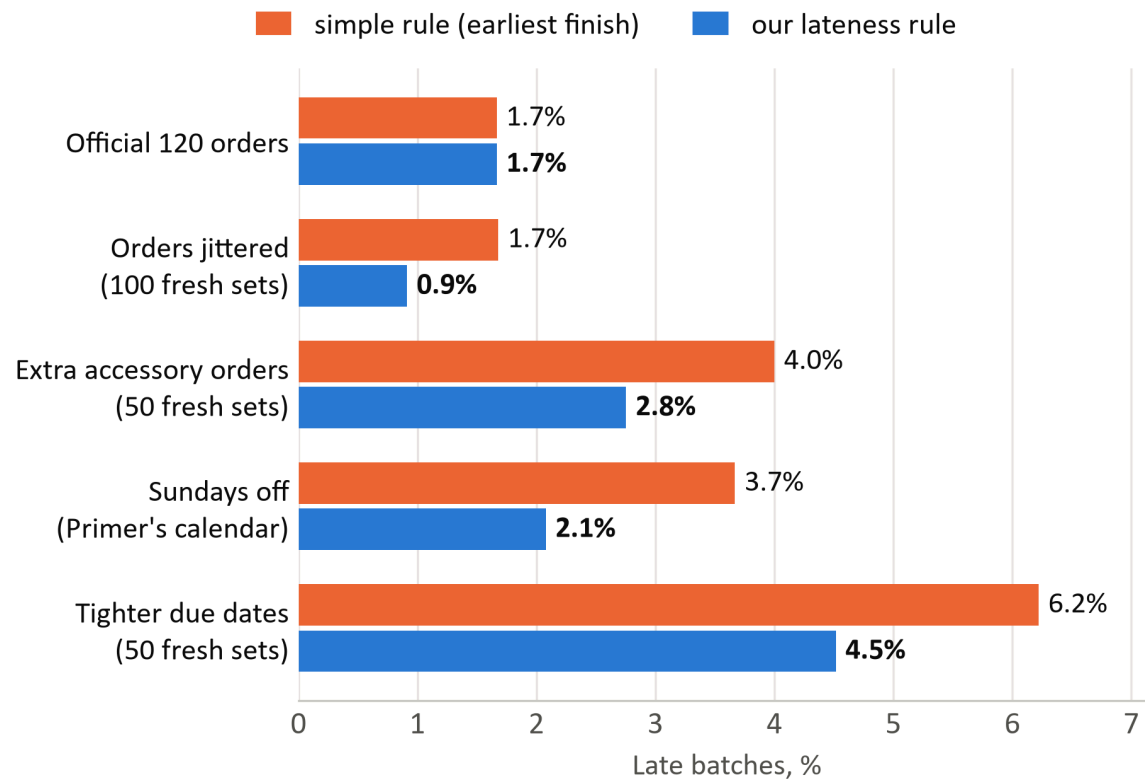
of “if reworked” dates hold (99.2% with a workshop closed). They are 1.8 days later on average.

SUNDAYS OFF, AS THE PRIMER SAYS · LATE BATCHES %

Policy	Seven-day week	Sundays off
random	34.2	50.2
earliest_finish (simple rule)	1.7	3.7
lateness rule (ours)	1.7	2.1
lateness rule, estimate knows Sundays	1.7	1.8

Same orders, same defect draws; only change: no work on Sundays. The allocators are not told. With the Sunday rule switched off the variant reproduces simulate.py exactly.

Appendix E · Our rule ties the simple rule on the official data, and wins when the data moves



The rule, in order: on time even if the batch is reworked; then the least flexible shop (a specialist before one that makes both categories); then earliest finish.

- **Why it helps:** earliest-finish spends the fast, flexible shops on batches that had slack; the next tight batch then has nowhere to go.
- **Official data:** a tie on late batches (1.7%), half the late days (0.09 against 0.18).
- **Fresh data:** better on 72 of 100 jittered order books, worse on 3.
- **Where the simple rule still wins:** mean turnaround, 8.3 against 9.3 days; the shock run, by one batch in ten.

Official: seeds 5105 to 5114. Fresh sets were generated after the rule was fixed (seeds 20000+), 3 seeds each. The allocators are not told about Sundays.

Appendix F · Why not the setting with 1.0% late? Its edge does not hold on fresh data

Both settings share steps 1 and 2 (rework-safe shops, specialist first) and differ only in the tie-break: earliest finish, or the latest finish that is still safe (use_slack). Paired runs on the same data and seeds; a sign test on how often use_slack is later or earlier.

Data	Runs	use_slack better / equal / worse on late %	Sign test p	Mean late % (ours / use_slack)	Mean turnaround, days
Official 120 orders, 10 seeds	10	8 / 2 / 0	0.008	1.67 / 1.00	9.3 / 14.0
Official, shock, 10 seeds	10	5 / 4 / 1	0.219	2.00 / 1.67	9.4 / 14.0
100 jittered order books	100	38 / 40 / 22	0.052	0.91 / 0.79	8.8 / 13.6
50 sets, tighter due dates	50	26 / 4 / 20	0.461	4.52 / 4.37	8.0 / 10.0
50 sets, extra accessory orders	50	23 / 11 / 16	0.337	2.75 / 2.69	9.8 / 14.5

Reading. use_slack is reliably better only on the official orders, which both settings were shaped on (8 of 10 seeds, $p = 0.008$). On 200 fresh order books the two are indistinguishable on lateness ($p \geq 0.05$). The robust differences are elsewhere: use_slack returns batches about 5 days later and costs about 9% less. The track names turnaround days as the business impact, so we declared the faster tie-break and keep use_slack as the cost-saving setting.